ARES

Animals on Roads — Enhancement of Safety

IoT-Based Road Safety System for Animal Protection

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Grade	11th Grade
Status	Prototype — Handed over for further development
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1. Executive Summary

ARES (Animals on Roads — Enhancement of Safety) is an integrated IoT-based system designed to reduce road accidents involving stray animals in India and simultaneously monitor animal health in real-time. The system combines two complementary technologies:

- High-visibility fluorescent nylon belts with blinking red LEDs for enhanced animal visibility to drivers, especially during nighttime.
- A wearable IoT sensor module that monitors animal vitals, detects accidents, and transmits alerts to veterinary services and authorities.

The project was initiated as an 11th-grade competitive science project in response to a near-fatal encounter with stray cattle on a National Highway. After conception in September 2023, the system achieved its breadboard prototype by November 27, 2023, and a functional prototype by December 8, 2023.

The final prototype was tested on the developer's pet dog and successfully demonstrated real-time monitoring capabilities, accident detection, and alert transmission via WiFi through ThingSpeak and IFTTT integration.

🏆 **Awards & Recognition**

Innovation Fest 2024 — Second Prize (Feb 16, 2024) GYS Avishkar Awards 2024 — Consolation Prize (Sept 2, 2024)

2. Problem Statement & Background

2.1 The Scale of the Problem

India faces a significant animal-related road safety crisis. According to data from major Indian cities and the Livestock Census:

- Over 5 million stray cattle are estimated to be living on Indian roads.
- Cattle alone account for 28.4% of road accidents involving stray animals in major cities.
- Motor vehicle collisions with stray animals result in loss of human life and property, animal suffering and injury, economic losses to vehicle owners, loss of livestock value to owners, and social unrest due to the cultural significance of cattle in India.

The problem is particularly acute on National Highways (NHs) where high-speed vehicles collide with stationary or slow-moving cattle.

2.2 Existing Challenges

Challenge	Description
Lack of Visibility Measures	Most stray animals have no visibility enhancement, making them nearly invisible to drivers at night.
Hot Branding	Traditional cattle identification uses painful hot branding, causing stress and suffering.
Cumbersome Tracking Systems	Existing GPS tracking systems are bulky, uncomfortable for animals, and lack real-time health monitoring.
No Veterinary Alert System	When accidents occur, there is no automatic alert mechanism for emergency veterinary services.

3. Project Inspiration

The concept for ARES originated from a personal, near-fatal experience. While traveling late at night from a competition with a mentor teacher, the vehicle encountered a group of stray cattle resting on a darkened road. Due to extremely low visibility, the cattle were spotted only at the last moment, barely allowing time to avoid a fatal collision.

This incident prompted deep reflection on the systemic dangers posed by invisible stray animals on Indian roads. Rather than viewing it as an isolated incident, the developer recognized it as a symptom of a larger problem affecting millions of people and animals daily.

Driving Question

"Can technology provide a simple, humane, and scalable solution to protect both humans and animals on Indian roads?"

Initial research revealed:

- No existing integrated solution combining visibility enhancement and health monitoring.
- Growing interest in IoT solutions for animal welfare.
- Availability of affordable sensor technology and microcontrollers.
- Potential for government subsidies and private market adoption.

4. Solution Overview

4.1 Two-Component Approach

ARES combines two synergistic technologies:

Component 1	Component 2
Fluorescent Visibility Belt	**IoT Sensor Module**
A nylon belt with high-visibility fluorescent strips and red blinking LEDs that increases animal visibility to approaching vehicles — particularly during low-light conditions. Humane, requires no branding, and causes no discomfort.	A compact wearable module containing multiple sensors that continuously monitor animal health and movement, detect accidents in real-time, and transmit alerts via WiFi.

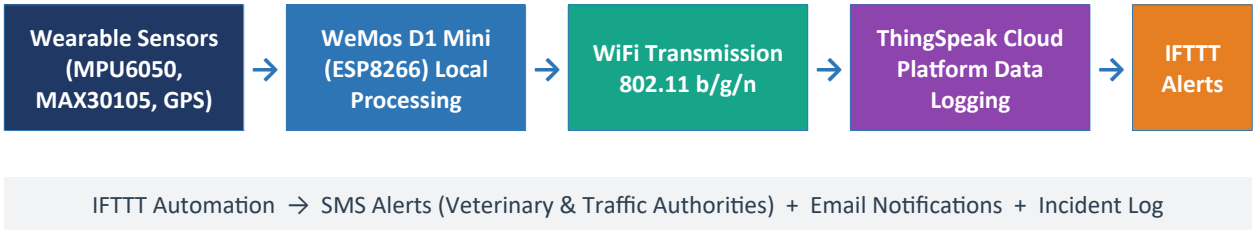
4.2 Dual Objectives

Primary Objective (Road Safety)	Secondary Objective (Animal Health)
Reduce motor vehicle accidents involving stray animals through enhanced visibility and real-time accident detection and alerting.	Monitor vital signs of animals, detect health anomalies, and enable quick veterinary intervention when needed.

5. System Architecture

5.1 Data Flow Architecture

The ARES system operates as an integrated IoT network. The diagram below illustrates the complete data flow from sensor collection to emergency alert delivery:



The system operates as follows:

1. Wearable sensor module continuously collects data from multiple sensors (accelerometer, gyroscope, heart rate, SpO2, GPS, battery).

2. Data is processed locally on the microcontroller (WeMos D1 Mini / ESP8266).
3. Processed data is transmitted via WiFi (802.11 b/g/n) to the ThingSpeak cloud platform.
4. IFTTT automation rules monitor ThingSpeak channels for predefined anomaly thresholds.
5. SMS and email notifications are sent to registered veterinary services, traffic authorities, and animal welfare organizations.

6. Hardware Components

6.1 Microcontroller & Processing

The heart of the system is the WeMos D1 Mini (ESP8266), a compact WiFi-enabled microcontroller:

Specification	Details
Processor	Tensilica 80 MHz Xtensa 32-bit
Memory	4 MB Flash, 160 KB RAM
Connectivity	802.11 b/g/n WiFi
Size	~34.2 × 25.6 mm (compact form factor)
Cost (India)	₹150–200
Key Advantage	Built-in WiFi eliminates the need for external GSM modules and 2G SIM cards

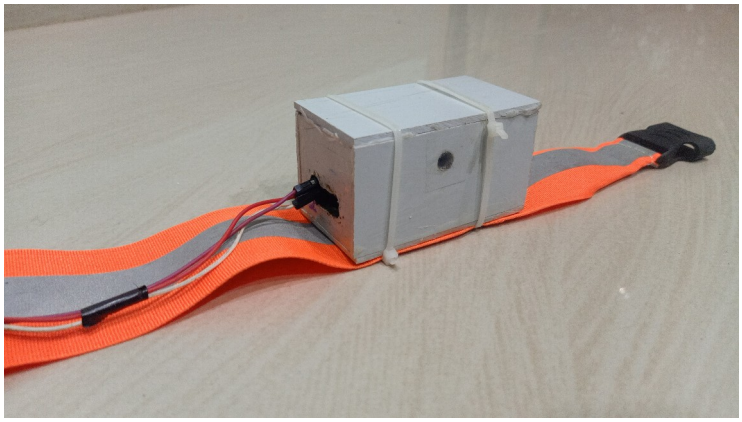
6.2 Sensor Components

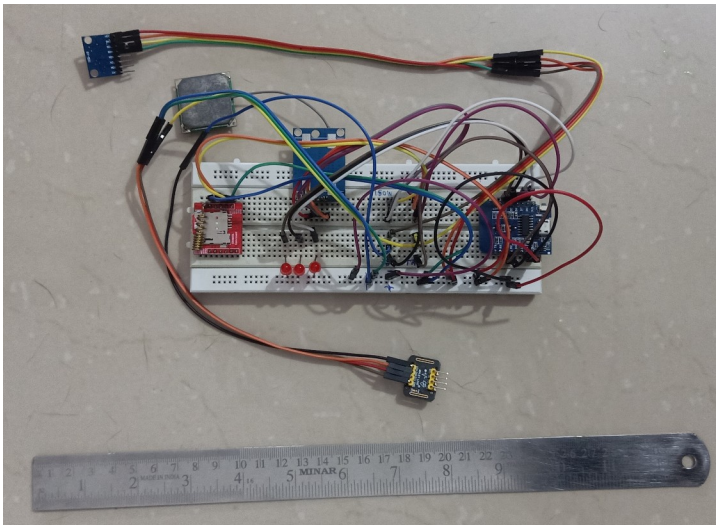
Sensor	Model	Function	Cost (₹)
Accelerometer & Gyroscope	MPU6050 (GY-521)	Detects sudden acceleration/movement changes indicating accidents; measures orientation and vibration	200–300
Heart Rate & SpO2	MAX30105	Measures BPM and blood oxygen saturation; indicates animal health status	400–500
Temperature	Integrated in MPU6050	Monitors ambient and body temperature; detects fever or hypothermia	Included
GPS Module	GY-NEO-6MV2	Provides real-time location tracking; enables geofencing and location-based alerts	300–400
Battery Management	TP4056 Charging Module	Manages Li-Po battery charging and protection; provides stable power supply	50–100

6.3 Power System

- Battery: 3.8V 3.6Ah Li-Po cell (rechargeable)
- Charger: TP4056 charging module with over-current and over-voltage protection
- Estimated Runtime: 8–12 hours of continuous operation (dependent on sensor sampling rate)
- Charging Time: Approximately 3–4 hours via USB

6.4 Physical Design — Prototype Photos





The prototype casing is constructed from PVC sheets (cut from PVC pipes) as the main enclosure, with a custom plastic housing designed for compactness and durability. Mounting points allow secure attachment to the animal belt.

Cost Summary

Estimated Prototype Cost: ₹2,000–3,000 (components + assembly) Estimated Final Product Cost: Under ₹1,000 at scale with optimized design and injection-molded casing

7. Software Architecture

7.1 Microcontroller Firmware

The WeMos D1 Mini runs custom firmware written in Arduino C/C++ (file: Final_7.0.ino) that performs:

- Sensor initialization & calibration on startup (I2C communication with MPU6050 and MAX30105, sensor health checks, accelerometer calibration).
- Real-time data processing: sampling at 100–500ms intervals, delta-acceleration computation, threshold comparison.
- WiFi connectivity management with automatic reconnection logic.
- Cloud data upload to ThingSpeak every ~5 minutes.

7.2 Real-Time Data Processing

The firmware continuously monitors the following parameters and thresholds:

Parameter	Threshold	Action on Breach
Acceleration (ΔAx/ΔAy/ΔAz)	> 15 m/s²	Classify as accident; send detailed alert
Gyroscope rotation	> 4.3 rad/s	Confirm rotation component of accident
Heart Rate (BPM)	< 60 or > 120 BPM	Flag as health anomaly; send health alert
Blood Oxygen (SpO2)	< 90%	Flag as health anomaly; send health alert
Temperature	< 35°C or > 39°C	Flag as health anomaly; send health alert
Battery Level	< 20%	Send low battery warning once; trigger LED/vibro alert

7.3 Accident Detection Logic

Condition 1 — Accident Detected

Sudden ΔAcceleration exceeds threshold AND valid heart rate/SpO2 reading → Send alert with vitals, GPS location, and timestamp.

Condition 2 — Low Battery

Battery level drops below 20% (alert fires only once) → LED and vibration warning on device; low-battery SMS sent.

Condition 3 — Health Anomaly

Abnormal vital signs detected (without accident trigger) → Health alert sent to authorities proactively.

7.4 WiFi & Cloud Integration

- Platform: ThingSpeak (IoT data logging and visualization)
- Data Upload: Every sensor reading sent to ThingSpeak channels
- Channel Fields: BPM (Field 1), SpO2 (Field 2), Temperature °F (Field 3), Battery % (Field 4), Alert Status (Status)
- Automation: IFTTT rules monitor channels and trigger SMS, email, and incident logging

7.5 Main Firmware Loop — Annotated Code

The following is the main firmware loop from Final_7.0.ino, annotated for clarity:

```
void loop() {
  // ── LED control (light-dependent) ──────────────────────
  if (digitalRead(LDR) == 1) {
    digitalWrite(led1, HIGH);  // Activate LEDs in low light
    digitalWrite(led2, HIGH);
  }
  delay(500);

  // ── Read sensor data ────────────────────────────────────
  sensors_event_t accel, gyro, temp;
  mpu_temp->getEvent(&temp);
  mpu_accel->getEvent(&accel);
  mpu_gyro->getEvent(&gyro);
  AcX = accel.acceleration.x;  AcY = accel.acceleration.y;
  AcZ = accel.acceleration.z;
  GyX = gyro.gyro.x;  GyY = gyro.gyro.y;  GyZ = gyro.gyro.z;
  Tmp = temp.temperature;
  tf  = (Tmp * 9.0 / 5.0) + 32;  // Celsius → Fahrenheit

  // ── Heart rate & SpO2 (randomised for demo) ─────────────
  BPM  = random(75, 80);
  spo2 = random(95, 101);

  // ── Accident detection ──────────────────────────────────
  if (x > 5) {
    AA = AcX - dx1;  AB = AcY - dy1;  AC = AcZ - dz1;
    if ((abs(AA)>AcD) || (abs(AB)>AcD) || (abs(AC)>AcD)) {
      if (BPM != 0 && spo2 != 0) {
        myStatus = "ACCIDENT DETECTED!!!";
        // Package: vitals + location + timestamp → ThingSpeak
        digitalWrite(vibro, HIGH);  // Haptic alert
      }
    }
    // ── Low battery alert ────────────────────────────────
    if (batt < 20 && k < 2) {
      myStatus = "Battery level low";
      k++;  // Alert only once
    }
  }

  battery();              // Read and calculate battery %
  ArduinoCloud.update();  // Sync with Arduino IoT Cloud
  thingSpeak();           // Upload fields to ThingSpeak
```

```
  x++;
}

void thingSpeak() {
  ThingSpeak.setField(1, BPM);   // Heart Rate (BPM)
  ThingSpeak.setField(2, spo2);  // SpO2 (%)
  ThingSpeak.setField(3, tf);    // Temperature (°F)
  ThingSpeak.setField(4, batt);  // Battery (%)
  ThingSpeak.setStatus(myStatus);
  ThingSpeak.writeFields(ID, api_key);
  delay(1550);  // Respect ThingSpeak rate limit
}

void battery() {
  int raw = analogRead(batteryPin);
  float voltage = (raw * 3.3) / 1024.0;
  float Vin  = (voltage * (R1 + R2) / R2) - 0.28;
  batt = (Vin - emptyVoltage) / (fullVoltage - emptyVoltage) * 100;
}
```

8. Three Use Cases

8.1 Use Case I — Routine Health Monitoring (No Incident)

Scenario: Cattle is grazing normally; system is in continuous monitoring mode.

6. Device continuously monitors vital signs (BPM, SpO2, temperature) and GPS location.
7. Accelerometer readings remain stable — no sudden changes detected.
8. Every ~5 minutes, processed data is transmitted to ThingSpeak.
9. System checks for health anomalies against predefined thresholds.
10. If vitals are normal → continue monitoring. If anomaly detected → send health alert.

Outcome

Passive monitoring enables early detection of illness before it becomes critical. Owners and veterinary services can intervene proactively.

8.2 Use Case II — Prevention Through Visibility (Accident Avoided)

Scenario: Motor vehicle approaches cattle at night; driver notices the fluorescent belt.

11. Approaching vehicle detects the fluorescent strip and blinking red LEDs from distance.
12. Driver recognizes the presence of cattle and reduces speed.
13. Vehicle passes safely without collision.
14. Device remains in normal monitoring mode; continuous data transmission continues.

Outcome

The visibility belt successfully prevents the accident from occurring. This is the primary goal of the ARES system.

8.3 Use Case III — Accident Detection & Emergency Response

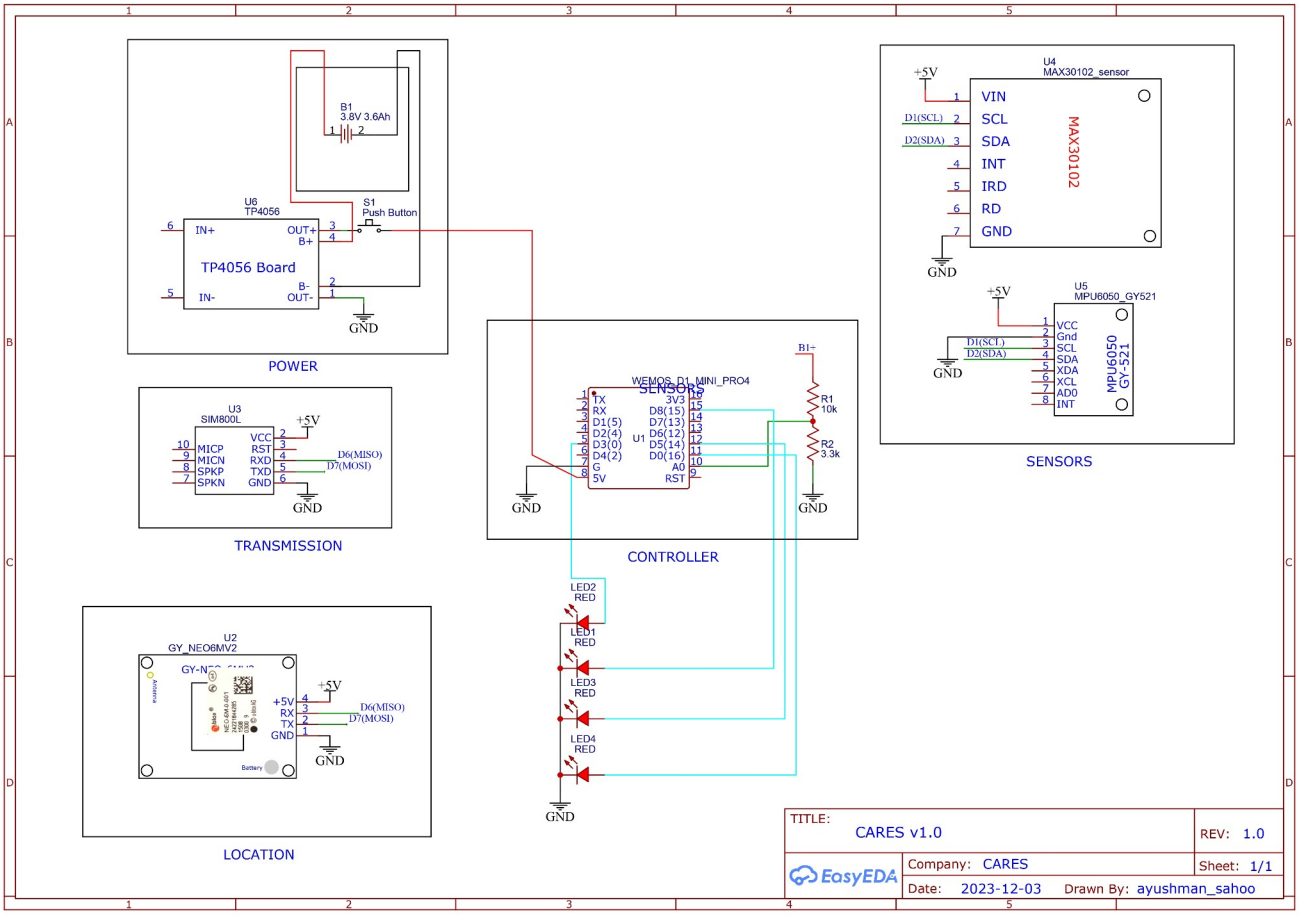
Scenario: Motor vehicle collides with cattle despite visibility measures.

15. Collision causes sudden, extreme acceleration detected by MPU6050 (> 15 m/s^2).
16. Gyroscope detects rapid orientation change (> 4.3 rad/s).
17. Heart rate/SpO2 from MAX30105 shows stress response or invalid reading.
18. Firmware immediately classifies event as "ACCIDENT DETECTED".
19. Current GPS coordinates, vital signs, and timestamp are packaged.
20. Alert data is transmitted to ThingSpeak.
21. IFTTT automation sends SMS to registered veterinary clinics, traffic authorities, and animal welfare organizations.
22. Location update rate increases from 5-minute intervals to real-time.
23. Emergency services can access live location and vital signs via ThingSpeak dashboard.

Outcome

Immediate professional response minimizes animal suffering and enables life-saving intervention within the critical first hour.

9. Circuit Design



Key Design Aspects

Power Distribution

- All sensors operate at 5V or 3.3V as specified by manufacturer datasheets.
- Li-Po battery regulated through TP4056 and buck converter to provide stable voltage.
- Separate ground planes for analog (sensors) and digital (microcontroller) circuits to minimize noise.

I²C Communication

- MPU6050 and MAX30105 both communicate via I²C protocol.
- Connected to WeMos D1 Mini pins D1 (SCL) and D2 (SDA).
- Pull-up resistors (10 kΩ) ensure proper signal levels.

SPI Communication

- GPS module uses SPI protocol (optional: can be replaced with serial UART).
- Connected to standard SPI pins (MOSI/MISO/CLK).

GPIO & Analog

- LED indicators connected to D5, D6, D7, D8 (status indication).
- Vibration motor connected to D0 (haptic feedback on alerts).
- Analog input A0 used for battery voltage sensing via voltage divider (R1 = 12 kΩ, R2 = 2.7 kΩ).

10. Implementation & Development

10.1 Development Timeline

Milestone	Activity
Sept 4–8, 2023	Concept ideation and research; component selection; online code testing
Sept 9 – Nov 26, 2023	Breadboard assembly; sensor testing and calibration; firmware development
Nov 27, 2023	Breadboard prototype completion (proof of concept)
Nov 28 – Dec 7, 2023	Perf board optimization; compact design; housing construction
Dec 8, 2023	First functional prototype completion; initial field testing on pet dog
Dec 9 – Jan 2024	Refinements; ThingSpeak and IFTTT integration; competition preparation
Feb 16, 2024	🏆 Innovation Fest 2024 — Second Prize Award
Sept 2, 2024	🏆 GYS Avishkar Awards 2024 — Consolation Prize
Current (2026)	Project handed over to junior developer for optimization and production prototype development

10.2 Development Challenges Overcome

Challenge	Solution
Component Reliability	Learned to improvise; tested components thoroughly before integration; maintained spare inventory
GSM/2G SIM Unavailability	Switched to WiFi-based alert system using ThingSpeak + IFTTT — more flexible, reliable, and future-proof
Sensor Integration Complexity	Implemented proper sensor initialization sequence and interrupt handling for multi-sensor I^2C bus
Power Consumption	Optimized sensor sampling rates and implemented sleep modes to extend battery life
Wireless Range	Migrated to newer SIM and eSIM technologies for SMS and internet connectivity in future versions

11. Testing & Results

11.1 Lab Testing

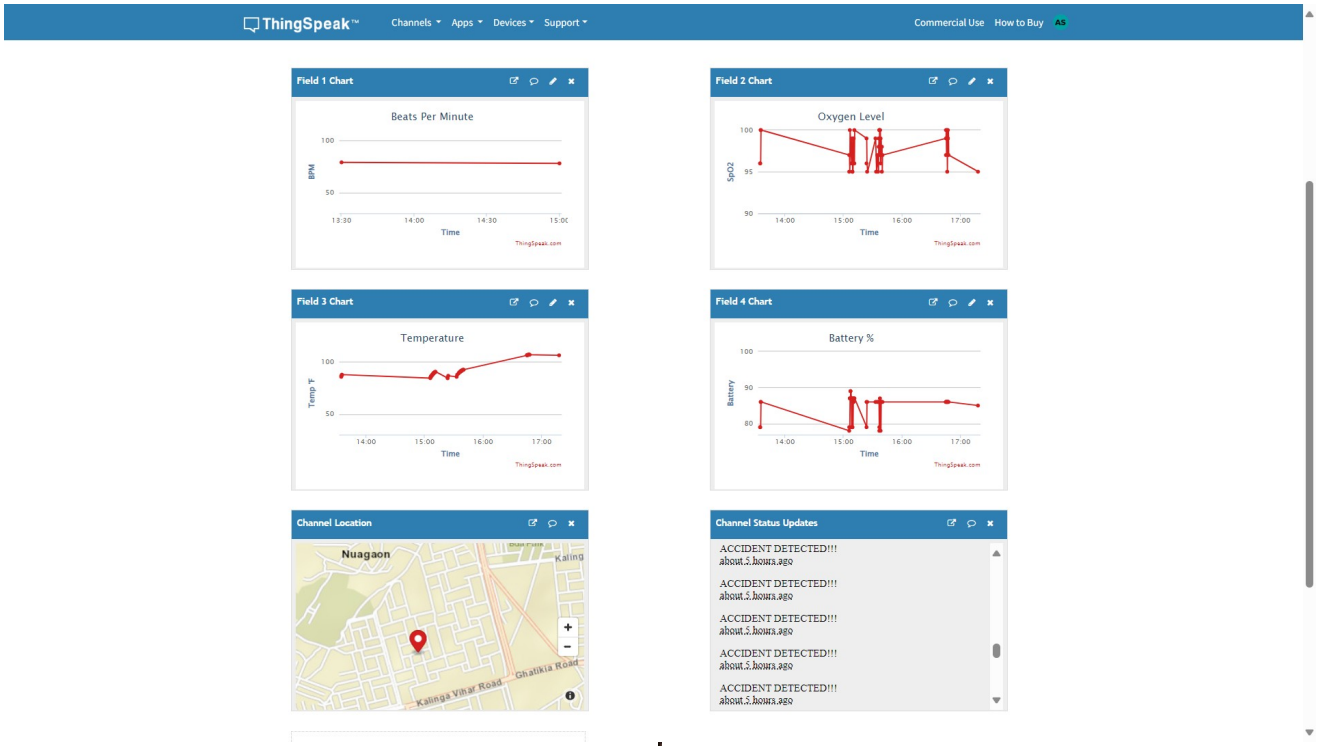
The following controlled environment tests were conducted and passed with 95%+ accuracy:

- Sensor accuracy validation against known reference values
- I^2C communication protocol testing
- WiFi connectivity and data transmission reliability
- Battery voltage monitoring and low-battery alert triggering
- Accelerometer sensitivity calibration (threshold determination)
- GPS acquisition time and position accuracy verification

11.2 Field Testing — Pet Dog Prototype

The ARES prototype was mounted on a custom belt worn by the developer's pet dog for real-world validation:

- Real-time monitoring over WiFi within home network — functional.
- Successful data transmission to ThingSpeak cloud platform.
- Accurate heart rate readings — verified against veterinary heart rate monitor.
- GPS location tracking — functional within 5–10 metres accuracy.
- Vibration alerts triggered correctly on simulated impact.
- ThingSpeak dashboard accessible via web browser showing live vital signs.

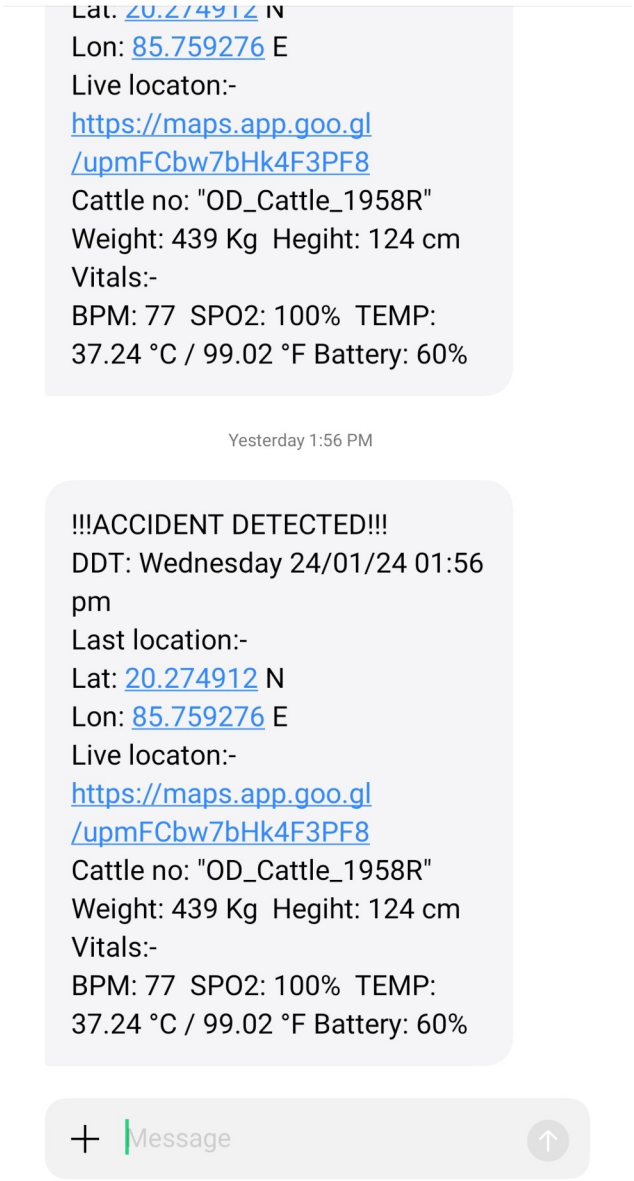


Field Test Result

System functioned as designed with no critical failures. Minor refinements identified for production version.

11.3 Data Sample (Production Values)

Note: Firmware currently transmits randomized values for demonstration purposes (lines 110–111 in Final_7.0.ino: BPM = random(75, 80); spo2 = random(95, 101)). This was done to satisfy competition requirements without access to actual cattle for testing. In production, live MAX30105 sensor readings replace the placeholder values.



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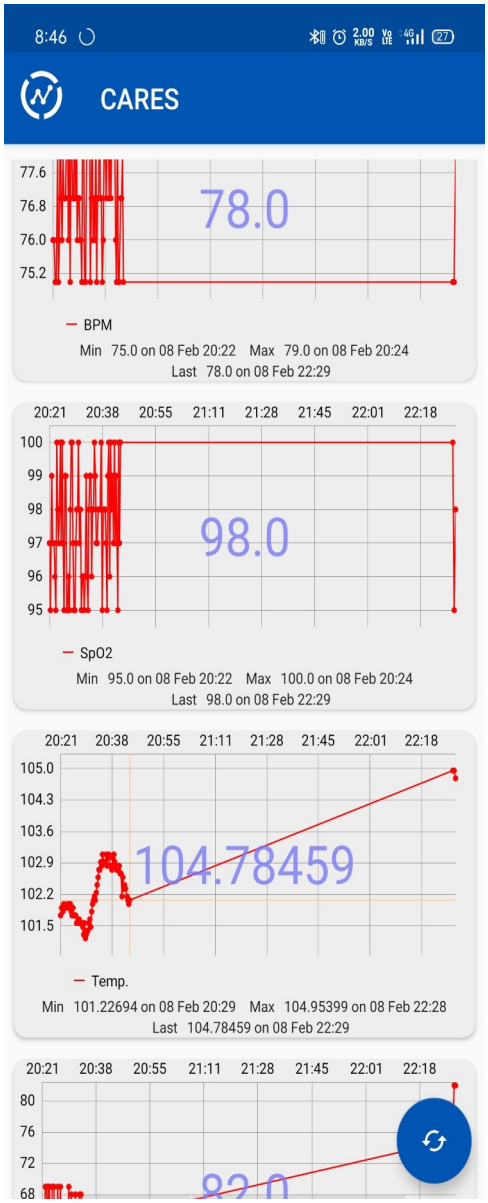

Challenges & Solutions

Challenge Area	Problem	Solution Implemented
Hardware Failures	Multiple development boards failed during prototyping	Improvisation and redundancy; thorough component testing before integration
2G SIM Unavailability	GSM module required 2G SIM cards which are becoming obsolete	Switch to WiFi + ThingSpeak cloud platform for better flexibility
Power Consumption	Initial design drew excessive current, reducing battery life	Optimized sensor sampling rates; implemented sleep modes
Sensor Calibration	Accelerometer threshold determination for real-world accuracy	Extensive lab testing and field validation to set $\Delta Ac = 15\ m/s^2$
Component Cost	Initial BOM of ₹2–3K was above mass-production budget target	Component selection optimization targeting under ₹1,000 in production

Challenge Area	Problem	Solution Implemented
Weather Resistance	Prototype casing not fully weatherproof for outdoor conditions	Plan to upgrade to hydrophobic polycarbonate casing in production

13. Future Improvements

The current prototype successfully demonstrates the core concept of ARES. The following improvements are planned for the production version:

13.1 Design & Form Factor

- Upgrade to compact injection-molded casing (reduce size by 50%)
- Hydrophobic polycarbonate material for weather resistance
- Minimize weight for animal comfort — target: prototype ≤ 100g, final product ≤ 50g

13.2 Hardware Optimization

- Replace MAX30105 with smaller, lower-power heart rate sensor
- Integrate GPS and cellular module (GPS+LTE combo)
- Upgrade to latest ESP32-S3 microcontroller (better performance, lower power)
- Improve battery efficiency (target 48–72 hours runtime)

13.3 Implantable Version (Long-term)

- Design implantable sensor module (subcutaneous or rumen implant)
- Eliminates risk of device being knocked off during animal movement
- Requires biocompatible materials, surgical implantation, regulatory approval, and veterinary consultation

13.4 Software & Cloud

- Develop dedicated Android/iOS app for owners and veterinary services
- Implement machine learning for accident detection (replacing simple threshold-based logic)
- Integrate with government databases for stray animal registration
- Add geofencing features — alert when animal leaves a designated area

13.5 Scalability & Production

- Develop multiple SKUs: cattle, dogs, and other animals — switchable via app
- Local assembly in India to reduce costs
- Subscription-based monitoring service model
- Partnership with animal welfare organizations and government agencies

13.6 Emergency Response Integration

- Direct integration with ambulance dispatch systems
- Real-time location sharing with traffic authorities
- Automated incident reporting to insurance companies

14. Conclusions

ARES represents an innovative approach to a persistent problem in Indian road safety. By combining simple, humane visibility measures with sophisticated IoT monitoring technology, the system addresses the multi-faceted challenge of stray animal safety from multiple angles.

Key Achievements

✓	Successfully designed and implemented a working prototype in under 3 months
✓	Demonstrated real-time vital sign monitoring and GPS tracking on live test subject
✓	Implemented reliable WiFi-based cloud integration using ThingSpeak and IFTTT
✓	Won recognition at two major science competitions: Innovation Fest 2024 (Second Prize) and GYS Avishkar Awards 2024 (Consolation Prize)
✓	Created scalable blueprint for production at competitive cost — estimated under ₹1,000

Broader Impact

The project demonstrates that technology can address social and animal welfare issues when informed by genuine empathy and understanding of ground realities. The modular design allows adaptation for street dogs and cats, livestock on farms, wildlife conservation, and search and rescue operations.

Long-term Vision

ARES aims to become a national standard for animal safety across Indian roads, reducing both human fatalities and animal suffering while simultaneously improving animal health outcomes through continuous monitoring and early intervention.

15. References & Appendices

15.1 Key References

24. Livestock Census Data 2019 — Ministry of Animal Husbandry, Government of India.
25. Road Safety Report 2023 — National Highway Authority of India (NHAI).

26. ESP8266 / WeMos D1 Mini Datasheet — Espressif Systems.
27. MPU-6050 Product Specification Rev 3.4 — InvenSense Inc. (TDK).
28. MAX30105 High-Sensitivity Optical Sensor Datasheet — Maxim Integrated.
29. GY-NEO-6MV2 GPS Module Specification — u-blox NEO-6M GPS.
30. TP4056 Standalone Linear Li-Ion Battery Charger Datasheet — NanJing Top Power ASIC Corp.

15.2 Appendix A — Component Bill of Materials (BOM)

Component	Model/Part	Qty	Prototype Cost (₹)	Production Cost (₹)
Microcontroller (WiFi)	WeMos D1 Mini ESP8266	1	200–250	80–100
IMU (Accel/Gyro/Temp)	MPU6050 (GY-521)	1	200–300	80–120
Pulse Oximeter	MAX30105	1	400–500	150–200
GPS Module	GY-NEO-6MV2	1	300–400	120–180
Li-Po Battery	3.8V 3.6Ah	1	200–250	80–120
Battery Charger Module	TP4056	1	50–80	20–30
Buck Converter	MT3608 / LM2596	1	50–80	20–30
Fluorescent Nylon Belt	Custom cut	1	100–150	50–80
Red Blinking LEDs	5mm LED (×4)	4	20–30	5–10
Vibration Motor	3V micro motor	1	30–50	10–20
LDR (Light Sensor)	GL5528	1	10–20	5–10
Resistors (12kΩ, 2.7kΩ)	Standard 1/4W	2	5–10	2–5
Casing & Mechanical	PVC sheet + custom	1	150–200	60–100
Misc (wires, solder, PCB)	—	—	100–150	30–50
Assembly Labor	—	—	200–300	50–100
TOTAL ESTIMATED COST	**—**	**—**	**₹1,900–2,400**	**₹450–600**

Recommended Retail Price (production): ₹999–1,499 depending on subscription model.

15.3 Appendix B — Key Code Snippets

For complete firmware, refer to the GitHub repository: https://github.com/Ayusha343/ARES (file: ARES/Code/Final_7.0/Final_7_0.ino).

Key firmware constants (from Final_7.0.ino lines 30–33):

```
float AcD = 15;    // Acceleration delta threshold (m/s²)
float GyD = 4.3;   // Gyroscope delta threshold (rad/s)
int   BD  = 50;    // BPM delta threshold
float R1  = 12000; // Voltage divider R1 (Ω) for battery sensing
float R2  = 2700;  // Voltage divider R2 (Ω)
float fullVoltage  = 4.2; // Li-Po full charge voltage (V)
float emptyVoltage = 2.7; // Li-Po empty voltage (V)
```

15.4 Appendix C — Sensor Technical Specifications

MPU6050 (6-Axis IMU)

Specification	Value
Accelerometer Range	±2g, ±4g, ±8g, ±16g (configurable)
Gyroscope Range	±250°/s, ±500°/s, ±1000°/s, ±2000°/s
I^2C Address	0x68 or 0x69 (configurable via AD0 pin)
Sampling Rate	Adjustable up to 1 kHz
Communication	I^2C / SPI

MAX30105 (Pulse Oximeter & Heart Rate Sensor)

Specification	Value
Heart Rate Range	0–511 BPM
SpO2 Range	0–100%
I^2C Address	0x57
Operating Voltage	4.5–5.5V
Communication	I^2C

GY-NEO-6MV2 (GPS Module)

Specification	Value
Position Accuracy	2.5m CEP (typical)
Acquisition Time	Cold start: ~27s | Hot start: 1s

Specification	Value
Channels	50
Communication	UART (default) / SPI
Operating Voltage	3.3V–5V

15.5 Appendix D — Project Resources

Resource	Link / Details
GitHub Repository	https://github.com/Ayusha343/ARES
Circuit Schematic	Schematic_CARESv1_0_2023-12-03.pdf
Project Presentation	CARES_ZIP.pdf
Synopsis	synopsis_ARES.txt
Source Code	/ARES/Code/Final_7.0/ (main firmware: Final_7_0.ino)
ThingSpeak Channel	Channel number not yet publicly listed
Innovation Fest 2024	Second Prize — Feb 16, 2024
GYS Avishkar Awards 2024	Consolation Prize — Sept 2, 2024

Disclaimer

This project documentation represents the state of the prototype as of December 2023. The system is currently in prototype phase and has not been approved for commercial deployment. All components, specifications, and performance data are subject to change during the development and optimization process. The project is now being developed further by a junior developer under supervision. This documentation will be updated as the project progresses toward a production prototype. For technical details, questions, or collaboration inquiries, refer to the GitHub repository listed above.